



SIMPLE, SPATIAL, SURVEY METHOD (S3M II) SUDAN

Summary



DATA COLLECTION:
AUGUST-DECEMBER 201
8

REPORT: JUNE 2019

Suggested citation:

Sudan S3M II (2018): Report of a Simple
Spatial Surveying Method S3M II in Sudan.

Federal Ministry of Health, Sudan.

2018 <http://www.fmoh.gov.sd>



Introduction

While Sudan is considered a lower middle-income country, large inequities persist throughout the country including with respect to the access to basic social services. Only with the ability to target investments in social services to the communities most in need will the available financial resources be effectively used to ensure improved service delivery necessarily for the development of a healthy and productive workforce and the further development of the country.

In addition, in response to both the onset of new and well as protracted emergencies, the ability to target service investments to those most in need is key to utilize the available humanitarian funding most effectively for the improved outcomes of the affected populations.

In 2013 Sudan conducted a national survey utilizing the Simple Spatial Survey Method (S3M), a large-scale survey method originally developed from the centric systematic area sampling (CSAS) coverage survey method (Myatt et al. 2005; Sadler et al. 2007) as a response to the widespread adoption of community management of acute malnutrition (CMAM) by ministries of health. From its initial application as a coverage survey method, S3M's potential for use in national-scale multi-indicator surveys was considered and explored.

Sudan supported the development of the method for this broader application through 2 state-level pilots in Gedaref and Sennar in 2012 which eventually led to the first national S3M survey in Sudan in 2013 (Sudan National S3M 2013) to measure a range of indicators (health, WASH and nutrition) over small geographical areas, giving results at locality level and lower. It provided comprehensive information at state level and lower that was used to target and re-align interventions.

Using the results from Sudan S3M I (2013) including mapping variation helped to identify areas of high need which allowed evidence-based targeting of interventions to most vulnerable populations, enabling improved cost-benefit in the current context of diminishing funding resources in Sudan. The S3M I contributed to planning of humanitarian and other responses to target the most vulnerable communities at locality level and below over the last several years in Sudan. For instance, the results were used for the mid-year revision of the 2014 Humanitarian Needs Overview (HNO) for all sectors enabling, for the first time, the use of locality level information for targeting responses, as well as acting as a catalyst for multi-sectoral programming. In addition, the S3M I survey results contributed to establishing the UNICEF programmatic shift to intervening in high-priority localities and it also helped the Government of Sudan to prioritize localities most in need.

In response to the need for updated, reliable and comprehensive data for health, nutrition, and WASH, it was decided to carry out the second S3M survey in 2018 (S3M II) which in addition to health, nutrition and WASH also included some child protection, education and food security indicators. Similarly, to the S3M I survey, the S3M II survey included data for smaller geographical areas within localities as well as from the national, state, and locality level for children and their mothers in Sudan. The results of the survey enables decision makers at all levels to identify the areas where the highest need is to allow evidence-based program planning and targeting for equity to enable a better and more effective use of scarce resources for humanitarian response and enhanced impact within Sudan as well as compare the results of the S3M-II to the first S3M carried out in 2013 to contribute to establishing trend data and identify uncovered areas in need. More specifically the specific objectives were as follows:

- To obtain data for health, nutrition, WASH, child protection, education and food security indicators for small areas (at sub-locality level) for children and their mothers in Sudan.

- To map results to show geographical areas of highest need and ‘hot-spots’.
- To provide evidence-base for prioritization of existing interventions, for health, nutrition, WASH, child protection, education and food security interventions, and to advocate for the situation of women and children in Sudan.
- To enable monitoring of trends at national state and locality level and to compare results between the two S3M surveys to monitor progress and change against baseline.

The S3M II survey was carried out in all 18 states of Sudan by the Federal Ministry of Health, UNICEF and other partners with technical support from Valid International and funding from European Union, DFID, AICS and others. Data from phase one states (North Darfur, East Darfur, West Kordofan, River Nile, Sennar, South Darfur, North Kordofan, Khartoum and Northern states) was collected in October 2018. Data from phase II states (White Nile, Kassala, Blue Nile, Central Darfur and West Darfur, Red Sea, South Kordofan and Gedaref states) was collected from November 2018 to January 2019.¹

Methodology

The survey used the Simple Spatial Survey Method (S3M), an area-based sampling methodology that uses settlement locations for sample selection. The survey was designed to be spatially representative of the whole country its smaller administrative units up to the locality level with the exception of few inaccessible areas².

An even distribution of primary sampling units (PSUs) (i.e., villages/city blocks) was selected from across the country. This approach was used as it is most suited to assessing indicators over wide areas to detect and map heterogeneity of indicators which is the primary objectives of this national survey (Gilbert 1987; Elliot et al. 2000; Pfeiffer 2008). PSUs (i.e. villages/city blocks) were selected based on their proximity to centroids of a hexagonal grid laid over the entire country. The resulting sample is a triangular irregular network (Pfeiffer 2008; E. H. Isaaks and Srivastava 1989). A variable density sampling approach (E. H. Isaaks and Srivastava 1989) was used to achieve a sample that draws a minimum number of PSUs from localities and from urban areas so that they can provide estimates for each of these areas with useful precision.

A sample of up to $n = 32$ mother and child pairs in $m = 3027$ PSUs was taken (see Figure 2.1). Across Sudan, a total of **93,882** households and **145,002** children below 5 years of age were surveyed.

¹ it should be noted that some areas were inaccessible due to insecurity. This affected parts of South Kordofan state (Elbouram, Omdorein and Heiban localities) and Abyei protocol area in West kordofan state. Areas not accessed for data collection are clearly shown on all maps. It should be noted that in some localities, some sample points were accessible when others were not. In this case results are presented for the locality with a note that it is only part of the locality. The area represented can be seen on the maps

² Inaccessible areas due to insecurity were excluded in this survey including: Three localities in South Kordofan state (namely Heiban, Um Dowrein and Al Buram), parts of East jebel Mara locality in South Darfur state, Abyei PCA area.

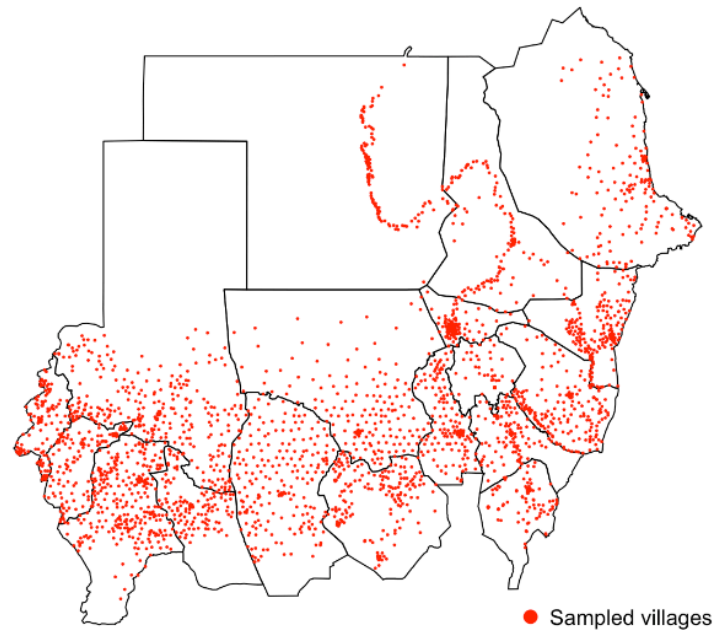


Figure 2.1: Map of selected PSUs

The *map-segment-sample* approach was used for within-community sampling of PSUs. In this approach, PSUs organised as ribbon/s of dwellings were sampled systematically while those organised as clusters of dwellings were sampled using a random walk EPI3 strategy (Bennett et al. 1994).

Preparations, data collection and analysis:

Planning of the S3M II survey and in particular, the timeframe of activities, was based on the previous experience of undertaking S3M-I but also influenced the inputs of stakeholders at federal and state level, particularly members of the S3M-II technical committee at federal level, which included WHO, WFP, the Ministry of Education (MOE), the Ministry of Security and Social Welfare (MOSSW) and the Ministry of Agriculture (food security directorate), besides the Ministry of Health and UNICEF.

FMoH and UNICEF, through the S3M II technical committee, regularly engaged with key stakeholders (one to two times per week during the planning phase) and ensured their involvement in the analysis stage. All technical committee members were invited to participate in the data analysis workshops. Other stakeholders - like the Central Bureau of Statistics (CBS) were consulted and informed of the progress achieved during each of the key stages (preparation, data collection and data analysis) while stakeholders such as donors were informed of progress. In addition, in every state, state level technical committees were formed and functioned to support the planning, data collection and analysis processes. The Ethical approval request was prepared by the FMOH and submitted to the ethics committee at the research department at the FMOH. Ethical approval was granted for both S3M-II and the nested micronutrients survey from FMOH. All other relevant approvals related to undertaking the survey were obtained including visas for international consultants supporting the survey, travel permits etc. Detailed maps for all states were obtained

through a successful partnership between UNICEF, FMOH and CBS. A joint workshop was undertaken in May 2018 followed by field visits in June 2018 to verify all coordinates was conducted (workshop hosted at the CBS). This was followed with a joint field visits to obtain and physically collect missing coordinates. Through UNICEF Valid International with Brixton Health were contracted and the organisations worked closely with FMOH and UNICEF throughout the survey as planned. It should be noted that the individuals included in the institutional contract were the experts who developed and later refined the S3M methodology as well as provided support to the wider Sudan team with the undertaking of the S3M-I survey in 2013. Initial list of the indicators to be included in the survey was developed through wide consultation with various sectors and stakeholders including government line ministries (MoH, MoE, MSSW, MoA) and UN agencies (UNICEF, WHO and WFP) through the S3M technical committee. The proposed list was further refined and revised by an external firm (Valid international organization) who also provided technical support throughout the survey. 226 indicators were collected and reported (see annex 1 for details).

Survey leads from UNICEF and MoH were identified and they further identified 9 UNICEF, 20 FMOH and 18 SMOH staff to supervise data collection and management. During Gezira pilot, the supervisors were trained to serve as data collectors to sharpen their skills to lead the data collection process in their respective states.. As a result, there were nine UNICEF, 20 Federal and 18 state level supervisors in addition to four supervisors from WHO who participated to oversee the collection of the micronutrient related data, as did their counterparts from the FMOH. Finally, a third-party ICT company was contracted by the FMOH to develop data collection digital tools and to provide ICT technical support and troubleshooting with regard to tablets and software issues, including presence of one ICT person in each state throughout the data collection. Sampling was carried out from 11 to 23 July 2018 by UNICEF and Ministry of Health staff in accordance with the S3M sampling methodology. This was based on the mapping of settlements across the entire country in villages and city blocks, carried out just prior to this (in 2.3.3 above), resulting in the assignment of correct GPS coordinates for more than 25,000 villages across the country.

Based on the mapping of settlements, distribution of primary sampling units was selected for each locality. This was done based on random sample selection using a sampling software designed to undertake S3M variable density sampling. The approach and selection were approved for their rigor and appropriateness by the external technical experts from Valid International. From the selected villages, consultations with state authorities including State Ministries of Health (SMOH), Humanitarian Aid Commission (HAC) and the National Intelligence and Security Service (NISS) were held to ensure the accessibility and security of villages for the survey teams and in the case of inaccessible villages, a replacement was made using the same sampling software. Further selection of households was done during the actual survey data collection; under the oversight of the external technical experts. Master training was carried-out in Gezira state (pilot state) in July-August 2018 to test tools including digital data collection tools, laboratory testing for micronutrients indicators and logistics and to train survey supervisors (through a Training of Trainers). Upon the identification of staff, all supervisors including the nine UNICEF supervisors, the four WHO supervisors, the 18 state supervisors and the 20 FMOH supervisors (including nine from Gezira state) received training from July 16 to July 25 on the following topics:

- S3M-II methodology.
- S3M-II questionnaire.
- Digital data collection tools (tablets).
- Micronutrients samples collection and storage.

- TORs of all personnel and groups.
- Anthropometric measurements standardisation.
- S3M-II monitoring tools.
- S3M-II logistic, data collection plans and quality assurance.
- Local calendar.
- S3M-II sampling including urban sampling.

In addition, 18 state nutrition directors received basic training on the S3M-II methodology, indicators, and the roles and responsibilities for committees involved. Further training for the supervisors for the micronutrient survey were done separately. This was followed by data collection in Gezira state which was carried-out by supervisors. Subsequent state level trainings were conducted prior to data collection for each state. Data from phase one states (North Darfur, East Darfur, West Kordofan, River Nile, Sennar, South Darfur, North Kordofan, Khartoum and Northern states) was collected in October 2018. Data from phase II states (White Nile, Kassala, Blue Nile, Central Darfur and West Darfur, Red Sea, South Kordofan and Gedaref states) was collected from November 2019 to January 2019. Close monitoring, supportive supervision and capacity-building to the Ministry of Health staff continued throughout the first phase. Technical assistance was provided from UNICEF S3M-II technical staff and Valid International throughout data collection.

Data Entry and analysis:

For the S3M-II survey it was decided to use tablets for digital data collection in real time to minimise data entry mistakes by having to translate written forms into digital form. To ensure the quality of data, all data entries were checked on a daily basis.

Data analysis was performed using R language for statistical computing (R Core Team 2018). The analytical approach for estimating indicators at locality, state and national level were as follows. Data analysis procedures were accounted for the sample design including the use of two-stage sample. Subjects were sampled from a small number of primary sampling units (PSUs). As this survey was **not** prior weighted, per-PSU sampling weights (the populations of the PSU) were applied during analysis.

For this survey, the *blocked weighted bootstrap*³ estimation approach was used:

- **Blocked** : The block corresponds to the PSU or cluster.
- **Weighted** : The sampling procedure for this survey does not use population proportional sampling to weight the sample prior to data collection as is done with SMART type surveys. This means that a posterior weighting procedure is required. The “roulette wheel” algorithm to weight (i.e. by population) the selection probability of PSUs in bootstrap replicates will be utilised.

A total of m PSUs are sampled *with-replacement* from the survey dataset where m is the number of PSUs in the survey sample. Individual records within each PSU are then sampled *with-replacement*. A total of n records are sampled *with-replacement* from each of the selected PSUs where n is the

³ The principal advantages of using a bootstrap estimator are:

- Bootstrap estimators work well with small sample sizes.
- The method is *non-parametric* and uses empirical rather than theoretical distributions. There are no assumptions of things like normality to worry about.
- The method allows estimation of the sampling distribution of almost any statistic using only simple computational methods.

number of individual records in a selected PSU. The resulting collection of records replicates the original survey in terms of both sample design and sample size. A large number of replicate surveys are taken (minimum of $r = 399$) replicate surveys but this can be changed). The required statistic (e.g. the mean of an indicator value) is applied to each replicate survey. The reported estimate consists of the 50th (point estimate), 2.5th (lower 95% confidence limit), and the 97.5th (upper 95% confidence limit) percentiles of the distribution of the statistic observed across all replicate surveys. The blocked weighted bootstrap procedure is outlined in annex 2. The indicator mapping creates a surface map of indicator values using spatial interpolation. There are various approaches and methods of spatial interpolation, the main differences are determined by the weights applied to the point dataset to estimate values at each of the unknown points of the surface map. For the Sudan S3M, spatial interpolation was performed using the inverse distance weighting (IDW) method. As the name implies, the IDW method uses weights that are inversely proportional to the distance of a point being estimated from the sampling point locations (E. Isaaks and Srivastava 1989; Diggle and Ribeiro 2007; Diggle 2013).

Results:

S3M II provided information on a range of indicators (health, nutrition, WASH, child protection, education and food security) over small geographical areas, giving results at locality level and lower. It has identified where the children who are most in need in Sudan are living ensuring evidence-based equity focused programming in Sudan.

Results in this summary are presented as:

1. Heat maps:

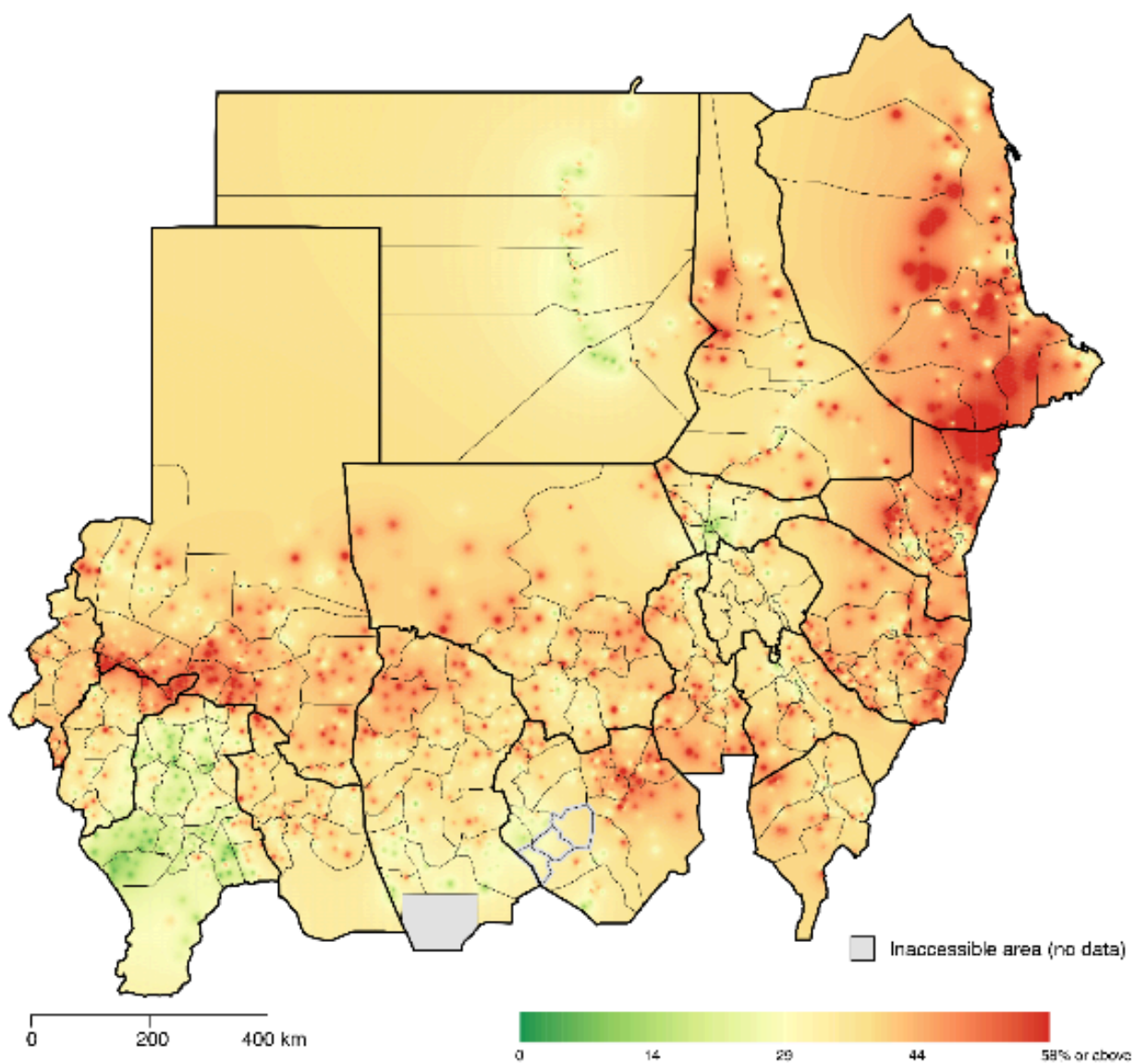
Generally, green colour on the maps indicates a 'good' situation, yellow indicates 'middle' and red indicates a 'poor' situation. The exact figure for the situation of the indicator in any given area/point on the map is equal to the figure that corresponds to same colour on the scale.

2. Results tables:

Locality level, state level and national estimates for all indicators are presented in to Tables.

Stunting: On a national level 36.6% of all children under the age of five are stunted. However, as with most of the indicators, national and state-level estimates mask huge disparities. For stunting 155 of the total 185 localities have a stunting rate classified as high (above 30 percent). There are pockets of very high stunting rates, found mostly in the Eastern and Darfur states of Red Sea, Kassala, Gedaref, North Darfur, East Darfur, Central Darfur and West Darfur with a high of 47 percent in North Darfur state.

Child : Height-for-age z-score < -2



⁴ Please note that Results maps presented in this report are read in light of the colour scale provided under each map

Table 1: stunting among children 6 – 59 months

Location	Child : HAZ < -2 Proportion			Child : HAZ between -3 and -2 Proportion			Child : HAZ < - 3 Proportion		
	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL
Northern State	28.84	26.17	31.64	18.97	15.80	22.52	9.97	8.27	11.60
River Nile State	37.39	35.17	39.40	21.93	19.24	24.56	15.42	13.49	17.08
Khartoum State	25.22	22.09	27.80	16.80	13.47	20.00	8.30	6.83	9.86
Al-Gazeera State	34.76	32.17	37.15	22.84	19.53	25.55	11.97	10.25	13.69
Sinar State	34.94	31.80	37.25	21.32	17.60	24.55	13.66	11.60	15.42
Blue Nile State	36.61	34.36	38.59	21.09	18.32	24.00	15.48	13.46	17.25
White Nile State	39.91	37.67	42.29	22.57	19.83	25.63	17.36	15.45	19.17
Red Sea State	46.03	43.00	49.45	23.90	19.33	27.71	22.33	19.40	24.95
Kassala State	43.83	40.67	46.63	24.48	20.33	28.27	19.45	16.92	21.87
Al-Gadarif State	45.52	43.21	47.38	22.84	20.17	25.70	22.57	20.68	24.28
North Kordofan State	39.40	37.78	41.94	24.08	21.44	26.79	15.36	13.85	17.48
South Kordofan State	36.80	34.41	38.99	22.07	19.43	24.71	14.87	13.51	16.50
West Kordofan State	36.12	34.28	37.91	21.97	19.83	24.23	14.09	12.68	15.71
North Darfur State	46.59	44.94	48.58	25.22	22.63	27.83	21.46	19.70	23.25
South Darfur State	23.85	22.21	25.91	17.65	15.46	19.76	6.21	5.46	7.30
East Darfur State	39.30	37.49	41.06	21.33	18.92	23.76	17.98	16.32	19.72
West Darfur State	42.40	39.95	44.13	22.79	19.93	25.12	19.52	17.89	20.92
Central Darfur State	43.67	40.62	46.34	22.29	18.31	25.81	21.32	19.02	23.66
Sudan	36.57	36.32	36.83	20.61	20.39	20.83	15.96	15.76	16.16

Wasting: Across Sudan on average 14.1% of children under the age of five years are wasted as measured as weight for height Z-scores (WHZ) of more than 2 standard deviations below the mean and 4.39% of children are severely wasted as measured as weight for height Z-scores (WHZ) of more than 3 standard deviations below the mean. The problem varies across the country. 56 of the 185 localities have a severe acute malnutrition rate (Severe wasting by weight-for-height z-score (WHZ)) that is classified as very critical (above 3 percent). Highest prevalence rates of SAM among under 5 children are found in Haya locality in Red Sea state (28 percent) and Shiaria locality in East Darfur (20 percent).

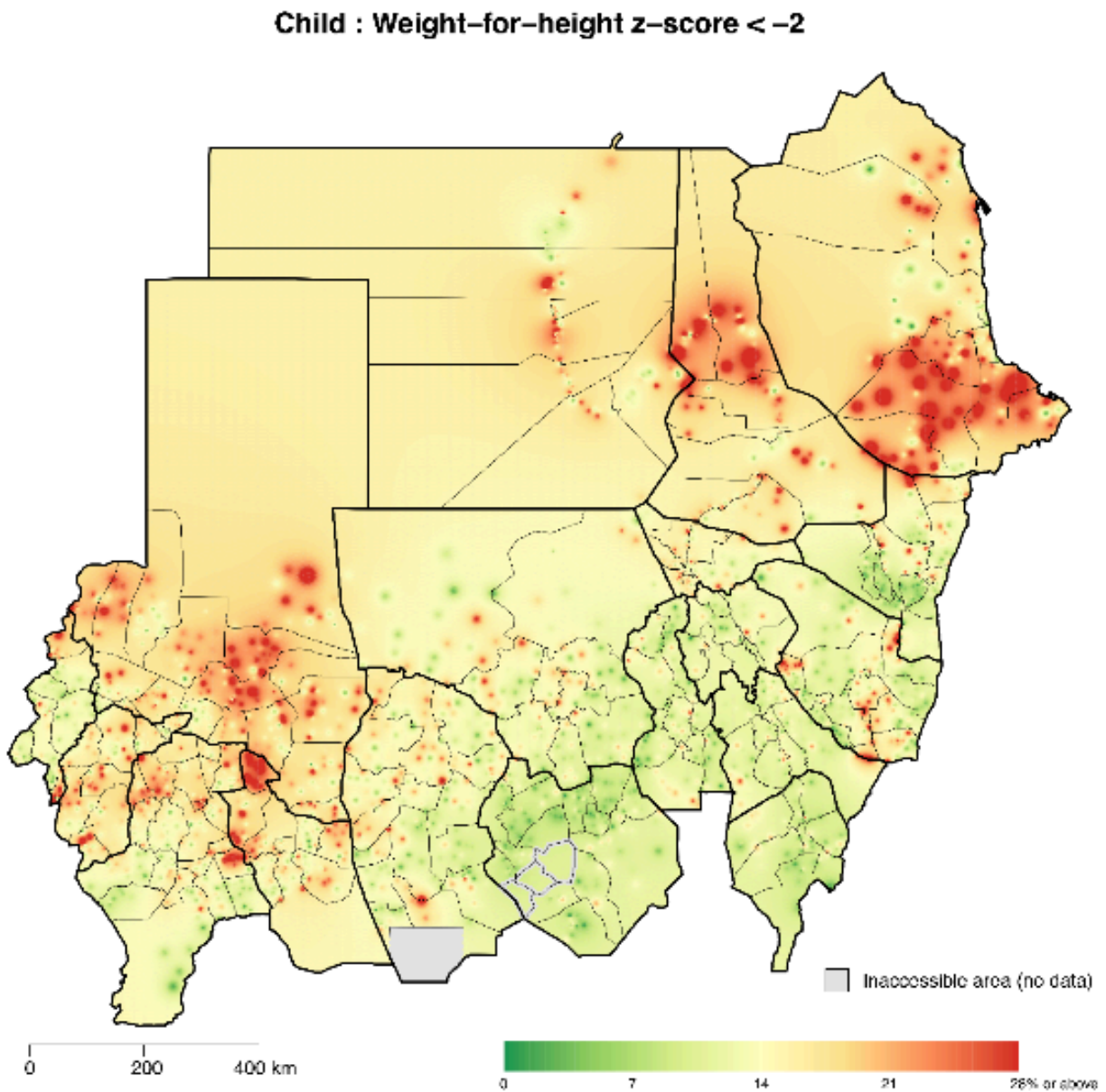
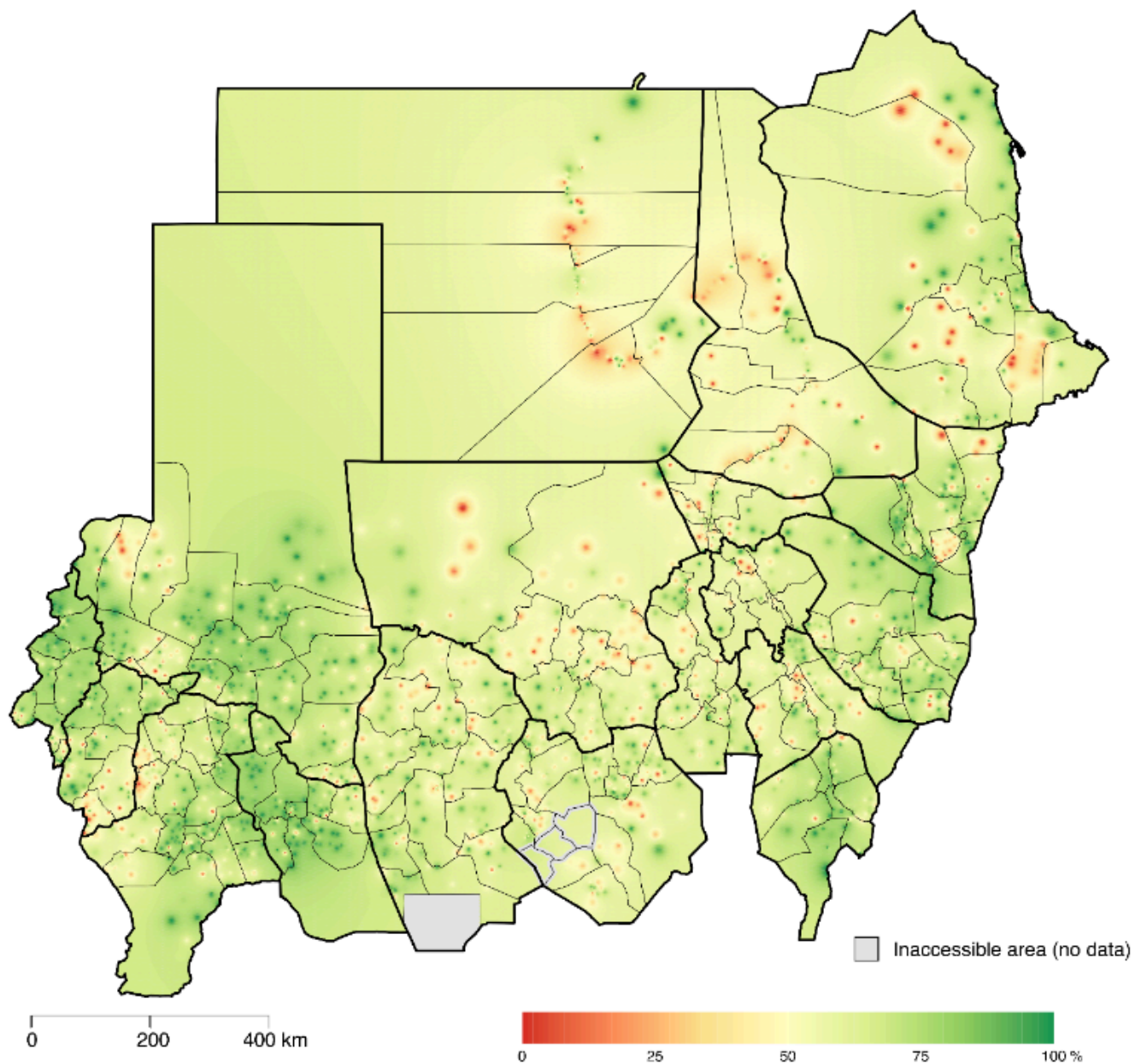


Table 2: Wasting by WHZ among children 6 – 59 months

Location	Child : WHZ < -2 Proportion			Child : WHZ between -3 and -2 Proportion			Child : WHZ < - 3 Proportion		
	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL
Northern State	17.69	15.48	19.62	13.94	11.65	15.95	3.70	2.83	4.73
River Nile State	17.25	15.24	19.09	13.06	10.78	15.17	4.18	3.16	5.21
Khartoum State	14.91	13.40	16.29	12.21	10.60	13.82	2.74	2.13	3.35
Al-Gazeera State	10.18	9.10	11.72	8.86	7.58	10.53	1.31	0.97	1.76
Sinar State	10.58	9.02	11.99	9.14	7.42	10.52	1.44	1.00	1.95
Blue Nile State	6.40	4.71	7.85	5.74	4.03	7.11	0.65	0.34	1.06
White Nile State	9.58	8.32	11.08	8.05	6.85	9.63	1.44	1.14	1.94
Red Sea State	17.84	15.63	20.44	12.88	10.34	16.37	4.93	3.69	6.59
Kassala State	8.12	6.98	9.17	7.14	5.98	8.33	0.93	0.68	1.23
Al-Gadarif State	10.69	9.41	12.77	8.65	7.11	10.80	2.05	1.54	2.94
North Kordofan State	11.46	10.29	13.01	10.18	8.81	11.70	1.32	1.01	1.68
South Kordofan State	8.13	7.01	9.32	7.28	6.05	8.35	0.93	0.68	1.19
West Kordofan State	11.10	9.85	12.65	9.68	8.37	11.30	1.41	1.09	1.75
North Darfur State	19.46	17.75	20.87	16.12	14.38	17.68	3.28	2.70	3.89
South Darfur State	16.43	15.19	17.88	13.60	12.36	15.10	2.82	2.36	3.28
East Darfur State	16.88	15.02	19.38	13.12	10.71	15.84	3.76	2.87	5.05
West Darfur State	9.61	8.30	11.17	8.15	6.80	9.76	1.41	1.03	1.94
Central Darfur State	15.28	13.50	17.09	12.08	10.29	14.15	3.11	2.46	3.99
Sudan	14.10	13.91	14.28	9.70	9.54	9.87	4.39	4.28	4.50

Exclusive breastfeeding: rates of exclusive breast feeding have improved across the country since the last survey with on average 61.5% of children aged 0-5 months being exclusively breast fed. This percentage is above 70 percent in 72 of the 185 localities. However rates remain very low in selected areas of the country as seen in the map below.

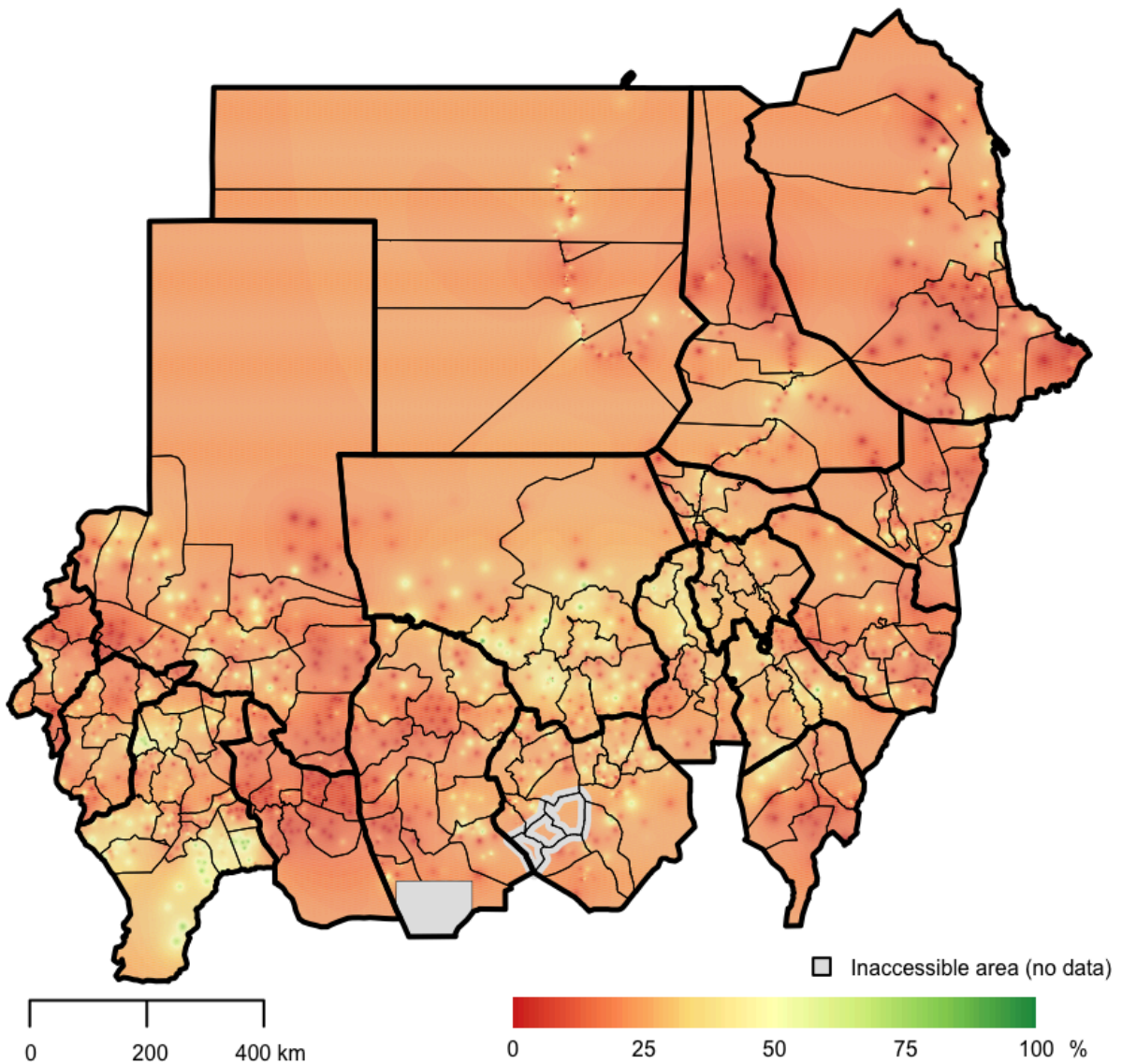
IYCF : Exclusive breastfeeding



Dietary Diversity:

Based on the results of the S3m II, the main challenge with infant and young child feeding in Sudan is the low age-appropriate dietary diversity of children 6-23 months old which is very across the country.

IYCF : Age-appropriate dietary diversity



Iodized salt: Nationally the use of adequately iodized salt is very low at only 4.67% on average. Only 3 localities across the country have more than 50 percent of households using iodised salt. Further, the Household use of iodised salt has declined since the last national survey.

Household : Salt iodization : Adequate (> 15 ppm)

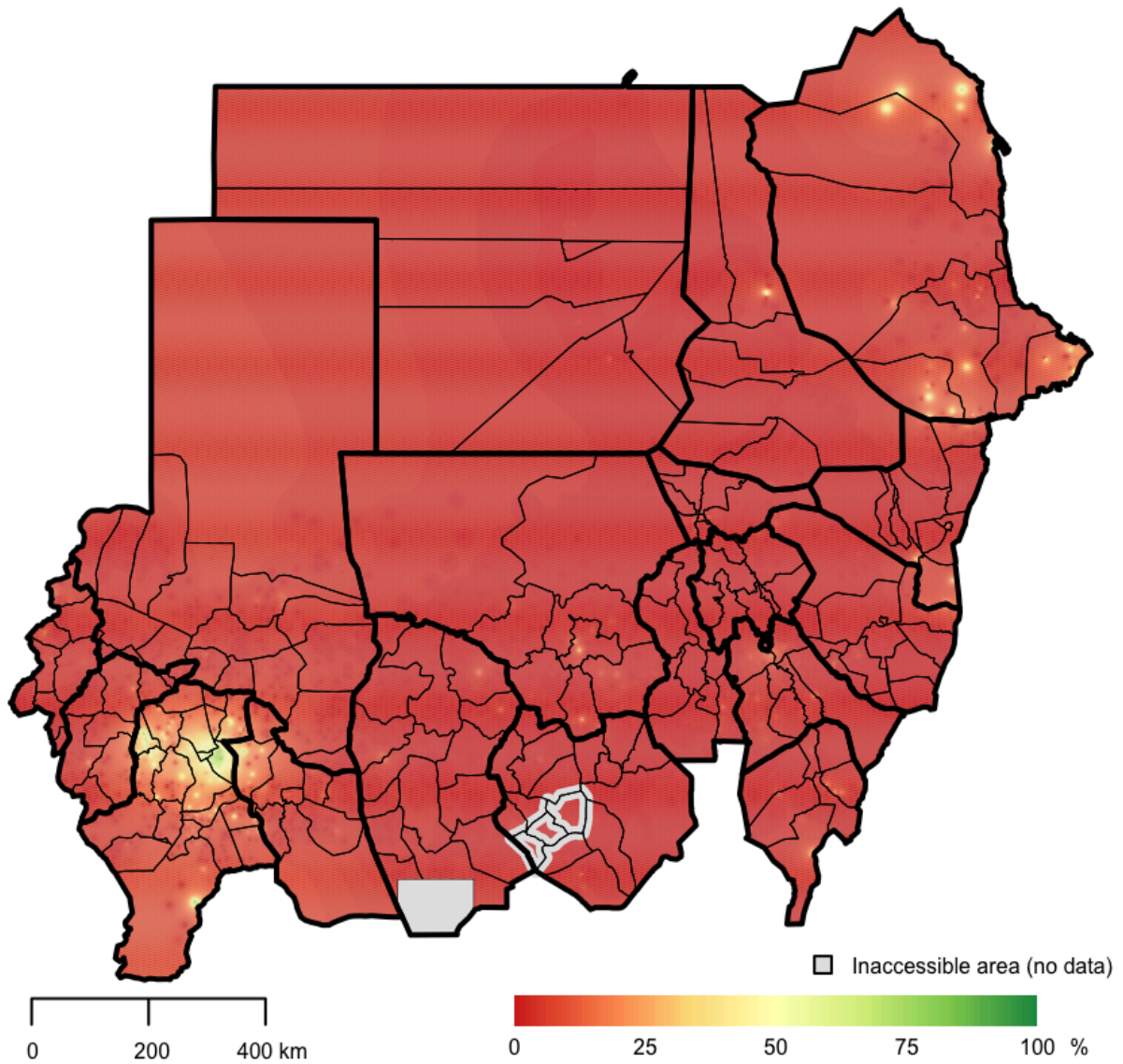
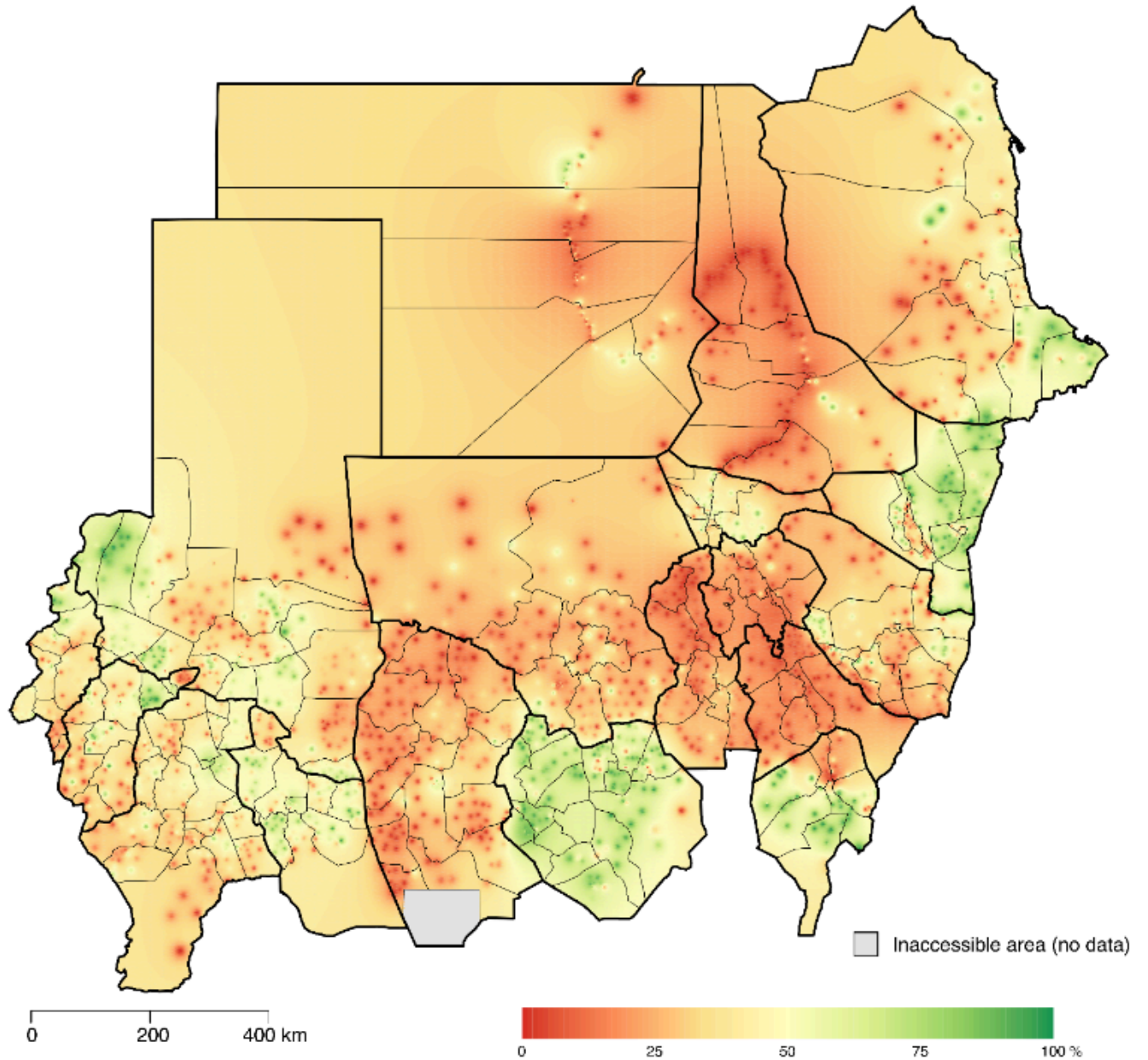


Table 3: Exclusive breastfeeding, dietary diversity and iodized saltLocation

location	IYCF: Exclusive breastfeeding			IYCF: Age-appropriate dietary diversity			Household: Salt iodization: Adequate (> 15 ppm)		
	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL
Northern State	53.04	46.40	60.19	16.76	13.47	19.98	0.45	0.14	0.84
River Nile State	45.64	40.19	50.78	30.94	26.23	35.23	1.63	0.48	2.85
Khartoum State	56.36	49.33	64.00	27.99	24.62	31.71	1.95	1.15	2.87
Al-Gazeera State	60.86	54.37	67.64	28.93	26.31	31.71	1.06	0.55	1.77
Sinar State	57.98	51.60	64.11	35.27	30.93	39.61	3.31	2.28	4.39
Blue Nile State	77.30	71.73	82.31	17.88	14.88	21.80	2.20	1.24	3.27
White Nile State	61.01	55.03	66.60	30.80	27.92	33.68	0.46	0.22	0.84
Red Sea State	65.02	59.20	70.50	25.81	22.15	29.55	4.29	2.75	6.36
Kassala State	75.08	69.95	79.53	25.10	21.73	27.97	3.66	2.25	5.45
Al-Gadarif State	63.09	57.31	68.49	18.55	15.87	21.06	0.34	0.13	0.66
North Kordofan State	55.45	50.80	60.47	33.92	30.40	37.22	1.19	0.65	1.78
South Kordofan State	61.12	55.81	65.59	23.58	21.36	25.82	0.48	0.27	0.82
West Kordofan State	59.26	54.80	63.65	20.31	18.06	22.45	0.78	0.43	1.18
North Darfur State	69.44	65.53	73.52	17.83	15.86	20.14	0.51	0.29	0.81
South Darfur State	66.50	63.02	70.11	30.42	27.91	32.99	38.09	35.05	41.37
East Darfur State	82.05	76.62	86.67	10.54	8.29	12.71	4.36	3.05	5.86
West Darfur State	80.85	76.39	84.85	18.91	15.87	22.00	1.34	0.69	2.35
Central Darfur State	69.65	63.94	74.64	17.00	14.68	19.55	1.72	1.14	2.50
Sudan	61.52	61.20	61.83	24.11	23.81	24.42	4.67	4.56	4.78

Vitamin A supplementation: The survey revealed low rates of Vitamin A supplementation. 52 localities recorded coverage of less than 75 percent, with a low of 0.0 percent in two localities in Central Darfur and South Kordofan States.

Child : Vitamin A supplementation coverage

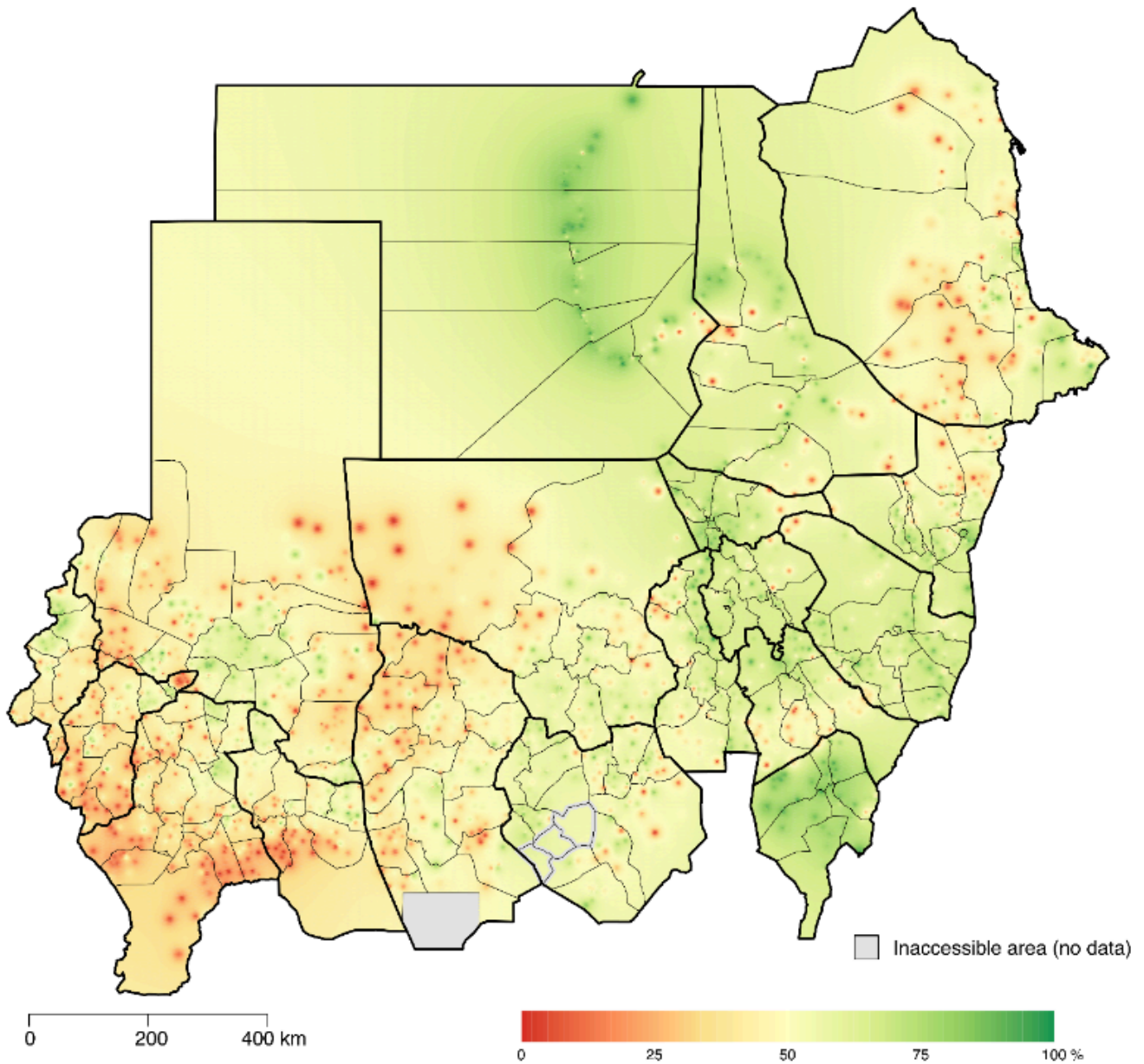


Epi coverage:

The Expanded Programme on Immunisation (EPI) forms the foundation of global public health action for child survival and disease prevention. Achieving high EPI coverage is important in ensuring programme effectiveness. In Sudan, a child will receive a vaccine against tuberculosis (BCG) at birth, three doses of pentavalent vaccine and three doses of oral polio vaccine before six months, and measles at 9 months.

Full EPI coverage of children less than 12 months old was at 25.95% (95% CI: 25.63-0.26%). Khartoum state had the highest full EPI vaccine coverage (40.57%) and North Kordofan state had the lowest full EPI vaccine coverage (15.89%).

EPI : Age-appropriate EPI coverage for children aged 0 – 24 months



Diarrhoea prevalence:

Period prevalence of diarrhoea is at 21.34% (95% CI: 21.13-0.22%). Al-Gazeera state had the highest period prevalence of diarrhoea and Central Darfur state the lowest.

Child : Period prevalence of diarrhoea

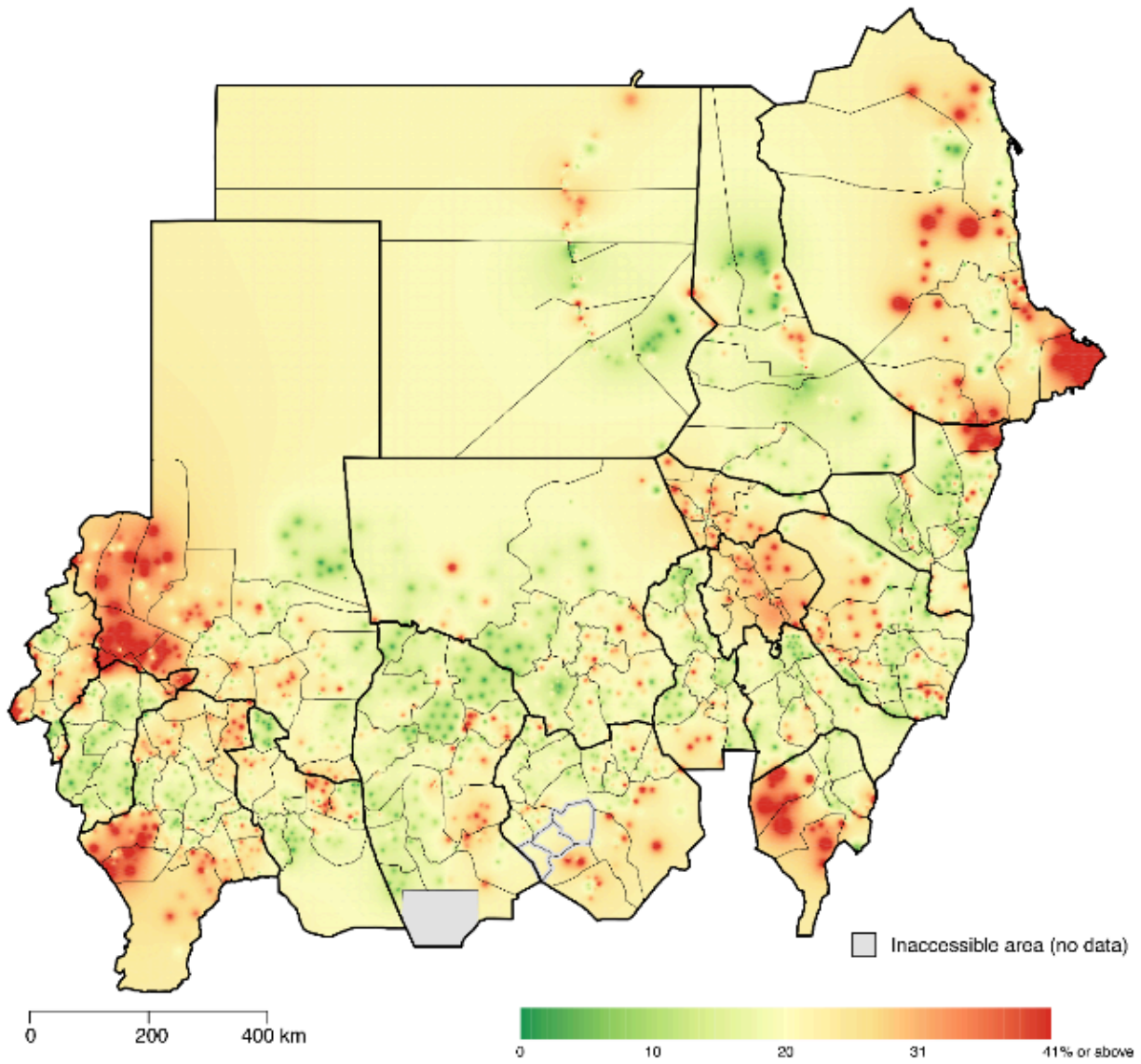


Table 4: Vitamin A supplementation, EPI coverage and period prevalence of diarrhoeaLocation

	Child : Vitamin A supplementation coverage	EPI : Full EPI coverage	Child : Period prevalence of diarrhoea
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	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL	Estimate	95%.LCL	95%.UCL
Northern State	26.92	22.63	31.65	29.04	25.37	32.96	16.27	13.74	18.86
River Nile State	8.03	5.44	10.62	27.09	23.78	31.33	13.58	11.69	15.21
Khartoum State	44.06	39.02	49.33	40.57	36.37	44.87	26.56	24.44	28.82
Al-Gazeera State	12.71	9.95	15.49	26.51	23.51	29.89	36.10	33.65	38.82
Sinar State	7.16	5.67	8.82	26.17	22.48	29.71	15.10	13.39	17.06
Blue Nile State	51.52	43.47	59.62	33.82	29.74	37.30	21.56	18.72	24.24
White Nile State	19.17	16.28	21.78	29.45	26.17	33.04	18.07	16.34	20.04
Red Sea State	58.24	53.83	62.62	29.06	25.45	32.21	27.53	24.89	30.11
Kassala State	67.73	63.38	72.09	25.93	23.46	28.24	18.36	16.23	20.66
Al-Gadarif State	37.44	32.76	41.31	30.77	27.88	33.63	20.15	17.94	22.27
North Kordofan State	21.89	18.53	25.29	15.89	13.36	18.08	16.30	14.52	17.93
South Kordofan State	79.95	76.62	82.66	22.08	19.42	24.65	21.84	19.98	23.59
West Kordofan State	19.03	16.46	22.19	19.94	17.45	22.68	17.73	15.96	19.54
North Darfur State	48.31	44.68	51.90	23.45	20.81	26.29	30.01	28.03	32.09
South Darfur State	55.20	52.54	58.02	18.77	16.43	20.75	26.13	24.94	27.72
East Darfur State	59.90	55.65	64.08	21.47	18.08	25.44	17.95	15.67	20.06
West Darfur State	35.49	31.39	39.58	22.22	19.24	25.49	29.40	26.72	32.04
Central Darfur State	49.68	44.06	55.72	19.52	16.73	22.47	12.92	11.32	15.31
Sudan	39.87	39.67	40.07	25.95	25.63	26.27	21.34	21.13	21.54

Remaining work and next steps:

A. Micronutrients biomarkers analysis:

S3M II also included the collection of blood samples to assess the micronutrients status for children aged 6 – 59 months and pregnant women as very little information on micronutrients status is available and the last national survey is more than 20 years old and hence has little relevance. Blood and urine samples were collected from 18 states to assess: Serum retinol, Urinary iodine, Serum ferritin, Transferrin binding receptor, and C-reactive protein in addition to the assessment of Haemoglobin by Hemocue (already analysed and reported). As funding was not available for the analysis earlier analysis has not yet started however UNICEF recently secured funds and subsequently efforts to undertake this remaining analysis has commenced.

Table 5: The time frame for the analysis and reporting on the micronutrient component is as follows:

Activity	Status	When	Responsible	Remarks
National lab avail necessary equipments and reagents for MN analysis	Ongoing	By End of October 2019	FMoH/WHO/UNICEF	Funds at FMoH need to be disbursed to move the procurement
Urine analysis for iodine	Done	By End of Oct 2019	FMoH _ National laboratory	
Analysis of Serum retinol	Not started	By End of Nov 2019	FMoH _ National laboratory	
Blood analysis for other biomarkers	5 states done 13 remaining	By End of Nov 2019	FMoH _ National laboratory	
Data entry	Started	By Mid - Dec 2019	FMoH _ National laboratory	
Matching MN data with survey data	Not started	By end of Dec 2019	FMoH _ National laboratory /Nutrition department	
MN data analysis	Not started	By end of January 2020	Valid International	Contracted by UNICEF
MN report writing and finalization	Not started	By end of March 2020	Valid International	Contracted by UNICEF
Launch of results and Dissemination of MNreports		April 2020	FMoH/UNICEF	
Finalization and printing of the S3M II main reports	Ongoing	Nov 2019	FMoH/UNICEF	
Documentation of S3M Sudan experience	Ongoing	March 2020	FMoH/UNICEF	

Financial support for the S3M II survey from UNICEF:

Besides the technical support for the S3M II survey through both its own personnel at both the field, Khartoum as well as additional international guidance from the international experts from Valid International, UNICEF have also supported the Ministry of health financially with the undertaking of the S3M II survey, primarily through the Federal Ministry of Health but also directly to the State Ministries of Health.

Table 6: The financial support in USD provided by UNICEF according to activity can be found below:

Costs	Total cost in USD
Maps	25,216.98
Anthropometric equipment and salt kits	24,299.09
Donor visibility and survey enumerator identification	15,634.53
Transport for teams and supervisors to sampling points for data collection	592,098.10
Support to Government of Sudan for mapping of settlements	54,638.00
Support to Government of Sudan for training and data collection including TOT training in Gezira state and state level trainings and data collection from all states	191,876.74
Support to Government for additional TOT training	62,789.00
Support to Government for additional data collection	170,269.00
Additional support for ICT	11,530.00
Additional financial support for data collection	63,158.00
Data analysis workshop	8,696.00
Report writing workshop	9,353.00
S3M II result launch	21,525.11
Support to the FMOH supervision for state level launches	4,934.22
Support to SMOH for state level launches	179,450.64
Support to the Government of Sudan for the Micronutrient survey including biomarker analysis and data analysis	127,672.22
Support to the Government of Sudan for the Micronutrient survey report writing and dissemination	13,000.00
Total	1,576,140.63

Besides the finalization of the remaining activities, UNICEF look forward to further collaboration with the Ministry of Health for the advocacy about the results and the use of the results in the planning and financing of interventions to collectively scale up the implementation of health and nutrition services for the people of Sudan.

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Annex 1

Indicators

The following indicators were assessed in Sudan S3M II.

Antenatal care indicators			
Indicator	Definition	Numerator	Denominator
At least one ANC visit	Percentage of PC reporting at least one ante-natal care visit during their last pregnancy	Number of principal carers reporting at least one ante-natal care visit during their last pregnancy	Total number of principal carers
Gestational age at first visit	Percentage of principal carers who had their first visit in the 1st /2nd /3rd trimester during their last pregnancy	Number of principal carers who had their first visit in the 1st /2nd /3rd trimester	Total number of principal carers
Four or more ANC visits	Percentage of principal carers reporting four or more ante-natal care visits during their last pregnancy	Number of principal carers reporting four or more ante-natal care visits during their last pregnancy	Total number of principal carers
ANC attended by trained personnel	Percentage of principal carers reporting their ANC visit was attended by a trained health care professional during their last pregnancy	Number of principal carers reporting their ANC visit was attended by a trained health care professional	Total number of principal carers
Fefol supplementation during pregnancy	Percentage of mothers reporting receiving iron and folic acid tablets (tablets shown) for one month / six months or more	Number of principal carers reporting receiving iron and folic acid tablets for 1 month / 6 months or more during their last pregnancy	Total number of principal carers
Coverage of Post-natal care	Percentage of principal carers who reported receiving a health check while in facility or at home following delivery, or a post-natal care visit within 2 days after the last delivery	Number of principal carers who reported receiving a health check while in facility or at home following delivery, or a post-natal care visit within 2 days after the last delivery	Total number of m women age 15-49
Night blindness during last pregnancy	Percentage of principal carers reporting difficulty seeing in the evening or night or suffering from night blindness during last pregnancy	Number of principal carers reporting difficulty seeing in the evening or night or suffering from night blindness during last pregnancy	Total number of principal carers

Health and nutrition information indicators			
Indicator	Definition	Numerator	Denominator
Knowledge of principal carers about malnutrition treatment services	Percentage of principal carers with knowledge about malnutrition treatment services	Number of principal carers with knowledge about malnutrition treatment services	Total number of principal carers
Knowledge of principal carer about Growth monitoring promotion (GMP)	Percentage of principal carers with knowledge about Growth monitoring promotion (GMP)	Number of principal carers with knowledge about Growth monitoring promotion (GMP)	Total number of principal carers
Preferable source of information/channel of health promotion messages	Percentage of principal carers reporting their preferred sources of information / health promotion (interpersonal, radio, TV, WhatsApp, Facebook etc.)	Number of principal carers reporting each preferences of different information channels.	Total number of principal caregivers
Preferred programme	Percentage of principal carers reporting Radio and TV as preferred sources, what is their preferred programme?	Number of principal carers reporting radio and TV as preferred sources	Total number of principal caregivers
Preferred time	Percentage of principal carers reporting radio and TV as preferred sources, who prefer listening / watching at morning / noon / evening / night	Number of principal carers reporting radio and TV as preferred sources, who prefer listening / watching at morning / noon / evening / night	Number of principal carers reporting radio and TV as preferred sources

Principal carer nutrition status indicator			
Indicator	Definition	Numerator	Denominator
Principal carer undernourished (MUAC <230 mm)	Percentage of principal carers with a MUAC less than 230 mm	Numbers of principal carers with a MUAC less than 230 mm	Total number of principal carers

Access to health facility and health personnel indicators			
Indicator	Definition	Numerator	Denominator
Access to Health services: nearby	Percentage of HH reported to have access to nearby HF	Number of HH reported to have access to HF within 5 km (1 hour walk)	Total number of HHs
Access to Health services: within 5km	Percentage of HH reported to have access to HF within 5 km (1 hour walk)	Number of HH reported to have access to HF within 5 km (1 hour walk)	Total number of HHs
Access to CHW	Percentage of HH reported to have access to CHW within 5 km (1 hour walk)	Number of HH reported to have access to CHW within 5 km (1 hour walk)	Total number of HHs

Salt iodization indicator			
Indicator	Definition	Numerator	Denominator
Adequately iodized salt	Percentage of households with a sample of salt tested as containing equal to or more than 15 parts per million of iodate	Numbers of households with a sample of salt tested as containing equal to or more than 15 parts per million of iodate	Total number of Household

Child protection indicators			
Indicator	Definition	Numerator	Denominator
Birth registration	Percentage of children 0-59 month	Number of children 0 - 59 months whose births are reported registered and/or have BC	Total number of children 0 - 59 months
Child marriage	Percentage of women aged 15-49 years who were married before age 15	Number of women age 15-49 years who were first married before age 15	Total number of women age 15-49 years
Child marriage	Percentage of women aged 15-49 years who are married	Number of women age 15-19 years who are married	Total number of women age 15-19 years
Female Genital Mutilation/cutting 1	Percentage of women aged 15-49 years who have undergone FGM	Number of women age 15-49 years who report to have undergone any form of FGM/C	Total number of women age 15-49 years

Child protection indicators			
Indicator	Definition	Numerator	Denominator
Female Genital Mutilation/cutting 2	Percentages of girls aged 0-14 years who have undergone FGM	Number of daughters age 0-14 years who have undergone any form of FGM/C, as reported by mothers age 15-49 years	Total number of daughters age 0-14 years of mothers age 15-49 years
Female Genital Mutilation/cutting 3	Percentages of girls aged 0-8 years who have undergone FGM	Number of daughters age 0-8 years who have undergone any form of FGM/C, as reported by mothers age 15-49 years	Total number of daughters age 0-8 years of mothers age 15-49 years

Recognition of danger signs indicator			
Indicator	Definition	Numerator	Denominator
Danger signs known by the principal carer for U 5 children	Average number of illness danger signs listed by principal carers	NA	NA

Principal carer diet indicators			
Indicator	Definition	Numerator	Denominator
PCs dietary diversity score	The average number of food groups eaten by PCs during the 24 hours before the survey	number of food groups eaten by PCs during the 24 hours before the survey	Total number of PCs
Principal carer undernourished (MUAC <230 mm)	Percentage of principal carers with a MUAC less than 230 mm	Numbers of principal carers with a MUAC less than 230 mm	Total number of principal carers
Child dietary diversity	Percentage of children consuming the appropriate number of food groups for their age during the 24 hours before the survey	Numbers of children consuming the appropriate number of food groups for their age during the 24 hours before the survey	Total number of children aged 0-59 Months
Meal frequency diagnostic	Percentage of children receiving meals the appropriate number of times for their age during the 24 hours before the survey	Numbers of children receiving meals the appropriate number of times for their age during the 24 hours before the survey	Total number of children aged 0-59 Months

Principal carer diet indicators			
Indicator	Definition	Numerator	Denominator
Introduction of complementary foods	Percentage of infants 6–8 months of age who receive solid, semi-solid or soft foods	Infants 6–8 months of age who received solid, semi-solid or soft foods during the previous day	Infants 6–8 months of age
Introduction of water during first six months	Percentage of children 0 – 5 months who received water in the past 24 hours	Number of children 0 – 5 months who received water in the past 24 hours	Total number of children aged 0-5 Months
Continuing breastfeeding	Percentage of children 6-24 months currently breastfeeding	Numbers of children 6-24 months currently breastfeeding	Total number of children aged 6-24 Months
Infant and Young child feeding Index	Average child feeding score among children 6-24 months	Numbers of child feeding score among children 6-24 months	Total number of children aged 6-24 Months

Child nutrition status indicators			
Indicator	Definition	Numerator	Denominator
Wasting Global (MUAC)	Percentage of children 6-59 months with a MUAC <125mm and or oedema	Numbers of children 6-59 months with a MUAC <125mm and or oedema	Total number of children aged 6-59 Months
Wasting MODERATE	Percentage of children 6-56 months with a MUAC <125 mm ==>115mm	Numbers of children 6-56 months with a MUAC <125 mm ==>115mm	Total number of children aged 6-59 Months
Wasting SEVERE (MUAC)	Percentage of children 6-59 months with a MUAC <115	Numbers of children 6-59 months with a MUAC <115	Total number of children aged 6-59 Months
Wasting Global (WHZ)	Percentage of children 6-59 months<-2 z-scores weight for height and/or oedema	Numbers of children 6-59 months<-2 z-scores weight for height and/or oedema	Total number of children aged 6-59 Months
Wasting MODERATE (WHZ)	Percentage of children 6-59 months<-2 z-scores for height and =-3 z-scores weight for height	Numbers of children 6-59 months<-2 z-scores for height and =-3 z-scores weight for height	Total number of children aged 6-59 Months
Wasting SEVERE (WHZ)	Percentage of children 6-59 months <-3 z-scores weight for height and/or oedema	Numbers of children 6-59 months <-3 z-scores	Total number of children aged 6-59 Months

Child nutrition status indicators			
Indicator	Definition	Numerator	Denominator
		weight for height and/or oedema	
Underweight GLOBAL	Percentage of children 6-59 months <-2 z-scores weight for age	Numbers of children 6-59 months <-2 z-scores weight for age	Total number of children aged 6-59 Months
Underweight MODERATE	Percentage of children 6-59 months <-2 z-scores weight for age and =>3 z-scores weight for age	Numbers of children 6-59 months <-2 z-scores weight for age and =>3 z-scores weight for age	Total number of children aged 6-59 Months
Underweight SEVERE	Percentage of children 6-59 months <-2 z-scores height for age	Numbers of children 6-59 months <-2 z-scores height for age	Total number of children aged 6-59 Months
Stunting/ stuntedness Global	Percentage of children 6-59 months <-2 z-scores height for age and =>-3z-scores height for age	Numbers of children 6-59 months <-2 z-scores height for age and =>-3z-scores height for age	Total number of children aged 6-59 Months
Stunting/stuntedness SEVERE	Percentage of children 6-59 months <-2 z-scores height for age and =-3 z-scores height for age	Numbers of children 6-59 months <-2 z-scores height for age and =-3 z-scores height for age	Total number of children aged 6-59 Months
Overweight / obesity	Percentage of children 6-59 months > 2 z-scores weight for height /age	Numbers of children 6-59 months > 2 z-scores weight for height /age	Total number of children aged 6-59 Months

Childhood illnesses and treatment seeking behaviour indicators			
Indicator	Definition	Numerator	Denominator
Period prevalence of diarrhea	Percentage of children 0-59 months who suffered from diarrhea in the 2 weeks prior to the survey	Number of children reported 0-59 months who suffered from diarrhea in the 2 weeks prior to the survey	Total number of children 0-59 months
Care-seeking for diarrhoea	Percentage of children 0-59 months for whom advice or treatment was sought from a health facility or provider during the last episode of diarrhea.	Number of children 0-59 months for whom advice or treatment was sought from a health facility or provider	Total number of children 0-59 months

Childhood illnesses and treatment seeking behaviour indicators			
Indicator	Definition	Numerator	Denominator
		during the last episode of diarrhea.	
Proper diarrhea treatment	Percentage of children 0-59 months who were treated during the last episode of diarrhea by using 1) ORS packets; 2) ORS packets with Zinc supplements; 3) Zinc supplements; and, 4) ORT (ORS + Home fluid)	Number of children 0-59 months who were treated during the last episode of diarrhea by using 1) ORS packets; 2) ORS packets with Zinc supplements; 3) Zinc supplements; and, 4) ORT (ORS + Home fluid)	Total number of children 0-59 months
Period prevalence of ARI	Percentage of children 0-59 months who suffered from cough with difficult breathing in the 2 weeks prior to the survey	Number of children 0-59 months who suffered from cough and/or difficult breathing in the 2 weeks prior to the survey	Total number of children 0-59 months
Period prevalence of fever	Percentage of children 0-59 months who suffered from fever in the 2 weeks prior to the survey	Number of children 0-59 months who suffered from fever in the 2 weeks prior to the survey	Total number of children 0-59 months
Antibiotic treatment for ARI (children with cough and difficult breathing)	Percentage of children 0-59 months who received antibiotic treatment during the last episode of cough with difficult breathing	Number of children 0 – 59 months who received antibiotic treatment during the last episode of cough with difficult breathing	Total number of children 0-59 months
Care-seeking for children with acute respiratory infection (ARI) symptoms	Percentage of children 0-59 months for whom advice or treatment was sought from a health facility or provider during the last episode of cough with difficult breathing	Number of children 0 – 59 months for whom advice or treatment was sought from a health facility or provider during the last episode of cough with difficult breathing	Total number of children 0-59 months

MUAC screening coverage indicator			
Indicator	Definition	Numerator	Denominator
MUAC screening coverage	During the past 3 months?	Numbers of children 6-59 months having their MUAC measured in the past three months	Total number of children aged 6-59 Months

Infant and young child feeding indicators			
Indicator	Definition	Numerator	Denominator
Recommended Infant and Young Child Feeding (IYCF)	Percentage of children aged 0-24 months receiving the recommended infant and young child feeding practices	Numbers of children aged 0-24 months receiving the recommended infant and young child feeding practices	Total number of children aged 0-24 Months
Exclusive Breastfeeding	Percentage of children 0-6 months taking breast milk only	Numbers of children 0-6 months taking breast milk only	Total number of children aged 0-6 Months
Infant and Young child feeding Index	Average child feeding score among children 6-24 months	Numbers of child feeding score among children 6-24 months	Total number of children aged 6-24 Months
Continuing breastfeeding	Percentage of children 6-24 months currently breastfeeding	Numbers of children 6-24 months currently breastfeeding	Total number of children aged 6-24 Months
Introduction of complementary foods	Percentage of infants 6–8 months of age who receive solid, semi-solid or soft foods	Infants 6–8 months of age who received solid, semi-solid or soft foods during the previous day	Infants 6–8 months of age
Meal frequency diagnostic	Percentage of children receiving meals the appropriate number of times for their age during the 24 hours before the survey	Numbers of children receiving meals the appropriate number of times for their age during the 24 hours before the survey	Total number of children aged 0-59 Months
Child dietary diversity	Percentage of children consuming the appropriate number of food groups for their age during the 24 hours before the survey	Numbers of children consuming the appropriate number of food groups for their age during the 24 hours before the survey	Total number of children aged 0-59 Months

Expenditure programme for immunisation coverage indicators			
Indicator	Definition	Numerator	Denominator
Measles vaccination coverage - single dose	Percentage of children 9-59 who received 1 dose of measles vaccination	Number of children 9-59 reported that received 1 dose of measles vaccination	Total number of children 9-59 months

Expenditure programme for immunisation coverage indicators			
Indicator	Definition	Numerator	Denominator
Measles vaccination coverage - two doses	Percentage of children who received 2 doses of measles vaccination	Number of children 9-59 reported that received 2 doses of measles vaccination	Total number of children 9-59 months
BCG coverage	Percentage of children 6-59 months vaccinated for BCG by card, recall or scar	Number of children reported that received vaccination for BCG by card, recall or scar	Total number of children 6-59 months
Pentavalent drop-out rate	Percentage of children ≥ 4 months vaccinated for PENTA-1 but not PENTA-3	Number of children ≥ 4 months reported that vaccinated for PENTA-1 but not PENTA-3	Total number of children 6-59 months
Age appropriate vaccination	Percentage of children 6-59 months who had received full appropriate vaccinations for their age	Number of children reported 6-59 months who had received full appropriate vaccinations for their age	Total number of children 6-59 months
Equity of immunization coverage	Percentage of localities which reach 80 percent of PENTA 3 coverage	Number of localities which reach 80 percent of PENTA 3 coverage	Total number of localities

Insecticide-treated nets coverage indicator			
Indicator	Definition	Numerator	Denominator
Child sleeps under an ITN	Percentage of children 6-59 months who slept under an ITN on the night before the survey	Number of children reported use of ITN on the night before the survey	Total number of children 6-59 months

Child health insurance coverage indicator			
Indicator	Definition	Numerator	Denominator
Health insurance coverage	Percentage of population covered by health insurance	Number of people covered by health insurance	Total population in sampled HHs

Child vitamin A supplementation coverage indicator			
Indicator	Definition	Numerator	Denominator
Vitamin A supplement received	Percentage of children 0-59 months who received Vitamin A supplementation in the 6 months prior to the survey	Numbers of children 0-59 months who received Vitamin A supplementation in the 6 months prior to the survey	Total number of children aged 0-59 Months

Anaemia prevalence indicators			
Indicator	Definition	Numerator	Denominator
Hemoglobin	Percentage of (age group) with HB level in (different ranges)	Number of (age group) with HB level in (different ranges)	Total Number of (age group)

Coping strategies index indicator			
Indicator	Definition	Numerator	Denominator
Coping Strategies	The Reduced Coping Strategy Index (rCSI) is used to assess the level of stress faced by a household due to a food shortage. It is measured by combining the frequency and severity of the food consumption-based strategies households are engaging in. It is calculated using the five standard strategies using a 7-day recall period.	NA	NA

Education indicators			
Indicator	Definition	Numerator	Denominator
Access to education	Percentage of children people above 4 years who ever attended school (disaggregated by gender)	Number of pupils above 4 years who attended school (M/F)	Total number of people above 4 years recorded on HH roster (M/F)
Access to basic Education (Net enrolment)	Enrolment of the official age group (6-13) for Basic Education expressed as a percentage of the corresponding population by gender	Number of pupils enrolled/attending basic education	Total number of children aged 6-13 years

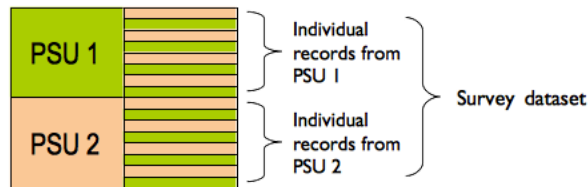
Education indicators			
Indicator	Definition	Numerator	Denominator
Reason for being out of school	Percentage of reasons for being out of school reported by PC	Numbers of pupils out-of-school reporting each reason	Total number of children out of school

Water, sanitation and hygiene indicators			
Indicator	Definition	Numerator	Denominator
Basic drinking water sources	Percentage of households reporting use of Basic Drinking Water sources which are those that have the potential to deliver safe water by nature of their design and construction, with collection time not more than 30 minutes for a roundtrip including queuing. They include 1) piped water; 2) boreholes or tube-wells; 3) protected dug wells/springs; 4) rainwater harvesting; 5) packaged (bottled water); and, 6) delivered (trucked from one of the above)	Number of households reporting use of Basic Drinking Water sources	Total number of households
Basic sanitation facilities	Percentage of households reporting use of Basic Sanitation Facilities which are designed to hygienically separate excreta from human contact and not shared with other households. They include 1) flush/pour flush to piped sewer system, septic tanks or pit latrines; 2) ventilated improved pit latrines; 3) composting toilets; and, 4) pit latrines with slabs	Number of households reporting use of Basic Sanitation Facilities	Total number of households
Practice of open defecation	Percentage of households which are not using any type of latrines and practicing open defecation which is identified as disposal of human feces in fields, forests, bushes, open bodies of water, beaches and other open spaces or with solid waste.	Number of households which are practicing open defecation	Total number of households
Handwashing practices	Percentage of principal carers with knowledge on the five handwashing critical times which are 1) after defecation; 2) after cleaning child who has defecated; 3) before eating; 4)	Number of principal carers with knowledge on each of five handwashing critical times which are	Total number of principal carers

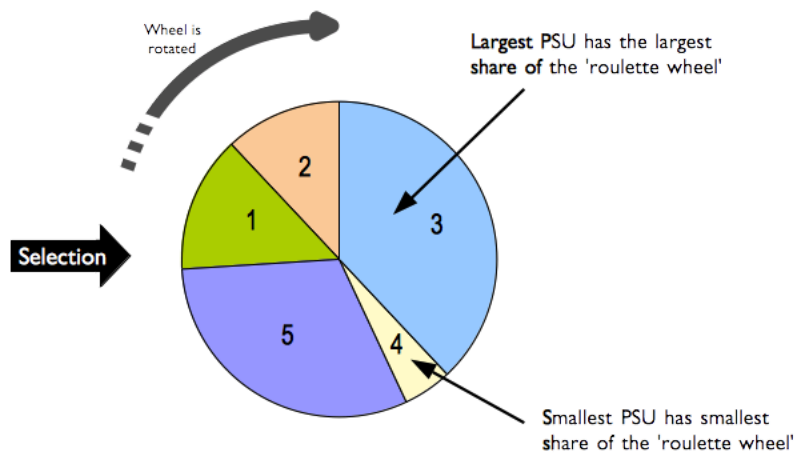
Water, sanitation and hygiene indicators			
Indicator	Definition	Numerator	Denominator
	before feeding child; and, 5) before preparing food and cooking		
Availability of handwashing facilities	Percentage of households with fixed or moving handwashing facility with soap and water available. Presence of ash instead of soap will indicate limited (not basic) handwashing facilities.	Number of households with fixed or moving handwashing facility with soap and water available.	Total number of households

Annex 2 steps in taking a blocked weighted bootstrap procedure

Data are collected using a two-stage cluster design:



PSUs are selected from the survey dataset *with-replacement* and with probability proportional to population size (PPS) using a *roulette wheel* algorithm :



PSUs are selected *with-replacement* and proportional to population size (PPS) :



Individual records are selected *with-replacement* from within each PSU to create a *replicate* survey. The estimator is applied to the replicate survey and the result recorded. This process is repeated many times. The estimate of the indicator value is made from the distribution of the results from each replicate survey which is the *empirical sampling distribution* of the indicator :

Estimate and 95% confidence interval from 50th, 2.5th, and 97.5th percentiles of the empirical (i.e. observed) sampling distribution of the indicator

